

ML4EE Test

Total Time: 75 Minutes; Total Marks: 45 (15%)

Type: Closed Book

Write all steps in derivations. Do not skip. State your assumptions clearly

Answer questions in order

Perceptron

You are given the following 2D linearly separable dataset with labels $y \in \{-1, +1\}$:

x_1	x_2	y
-3	2	+1
-1	1	+1
-1	-1	-1
2	2	-1
1	-1	-1

Initialize $\mathbf{w} = [-3, 2]^\top$, $b = 1$.

- (a) Run **two full epochs** processing points in the order listed. For each point in each epoch show: current $\mathbf{w}$ and b , the computed activation $\mathbf{w}^\top \mathbf{x}_i + b$, whether a misclassification occurs, and the updated $\mathbf{w}$ and b if so. (6 marks)
- (b) After your updates, write the equation of the final decision boundary and describe geometrically which points lie on which side. (2 marks)
- (c) Does the algorithm converge within two epochs? If not, state what convergence would look like, and identify which point is hardest to classify given the geometry. (2 marks)

Neural Network

Consider a neural network with input $\mathbf{x} = [1, 0]^\top$, two hidden layers with **3 units** and **2 units** respectively (all sigmoid activation), and a single output unit (sigmoid activation). The network uses the **Sum of Squared Errors (SSE)** loss function:

$$\mathcal{L} = \frac{1}{2}(\hat{y} - y)^2$$

where the true label is $y = 0.3$. The initial weights and biases are:

$$\begin{aligned}\mathbf{W}^{(1)} &= \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \\ 0.5 & 0.6 \end{bmatrix}, \quad \mathbf{b}^{(1)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \mathbf{W}^{(2)} &= \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.6 \end{bmatrix}, \quad \mathbf{b}^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \mathbf{W}^{(3)} &= [0.7 \quad 0.8], \quad b^{(3)} = 0\end{aligned}$$

- (a) Perform a complete **forward pass**. Compute the pre-activation values $(\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \mathbf{z}^{(3)})$ and activation values $(\mathbf{a}^{(1)}, \mathbf{a}^{(2)}, \hat{y})$ for every node in the network, and calculate the final loss $\mathcal{L}$. (5 marks)
- (b) Perform a complete **backward pass**. Compute the error signals $\delta^{(3)}$, $\delta^{(2)}$, and $\delta^{(1)}$. Defining: $\delta_j^{(l)} = \frac{\partial \mathcal{L}}{\partial z_j^{(l)}}$. Then, find the partial derivatives of the loss with respect to **all** weight matrices: $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}}$, $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(2)}}$, and $\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}}$. Recall: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. (6 marks)

- (c) Using a learning rate $\eta = 0.1$, write the updated value of the output weight matrix $\mathbf{W}^{(3)}$ and the first row of the first hidden layer weight matrix $\mathbf{W}^{(1)}$ after one gradient descent step. (2 marks)

Gradient Descent

You are minimizing $\mathcal{L}(w) = w^2 - 8w + 15$ using gradient descent. Initialize $w_0 = 7$, learning rate $\eta = 0.05$.

- (a) Compute $\frac{d\mathcal{L}}{dw}$ analytically. Run **5 steps** of batch gradient descent. For each step show w_t , $\left.\frac{d\mathcal{L}}{dw}\right|_{w_t}$, and w_{t+1} . What is the true minimum and how close are you after 3 steps? (5 marks)
- (b) Apply **Adagrad** defined as follows: $v_0 = 0$:

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{\sum_{\tau=1}^t g_\tau^2}} g_t \quad g_t = \nabla \mathcal{L}(w_t)$$

Run 3 steps. Does it attain a better minima as compared to vanilla GD at the same η ? Show numerically. (3 marks)

CNNs

- Consider a CNN that takes in 32×32 grayscale images and has a single convolution layer with three 5×5 convolution filters (without boundary padding). This is followed by a fully connected layer of size 32 and a single output node making a binary prediction.
 - What is the size of the output image after the filter is applied to the input image? (1 marks)
 - If boundary padding of size $p = 1$ is applied to the image, what is the size of the output image. (1 marks)
 - Calculate the speed up gained for forward-pass with the above CNN architecture (without padding) vs. a fully connected MLP (with a hidden layer of size 1024)? Only account for the multiplication and not the addition operations. (4 marks)

Proof

Q1. Illustrate the benefits of using Backpropagation algorithm over the standard way of calculating gradients in an MLP. Give an estimate of the speed up achieved by using the Backprop algorithm. (5 marks)

Conceptual Questions

- Q1. In a hard-margin SVM, only the support vectors determine the decision boundary. If you removed all non-support-vector points from the training set and retrained, would the boundary change? Why? (1 mark)
- Q2. Suggest 3 ways to perform regularization in a Feed Forward Neural network with justifications for each. Use ideas taught previously. (2 marks)